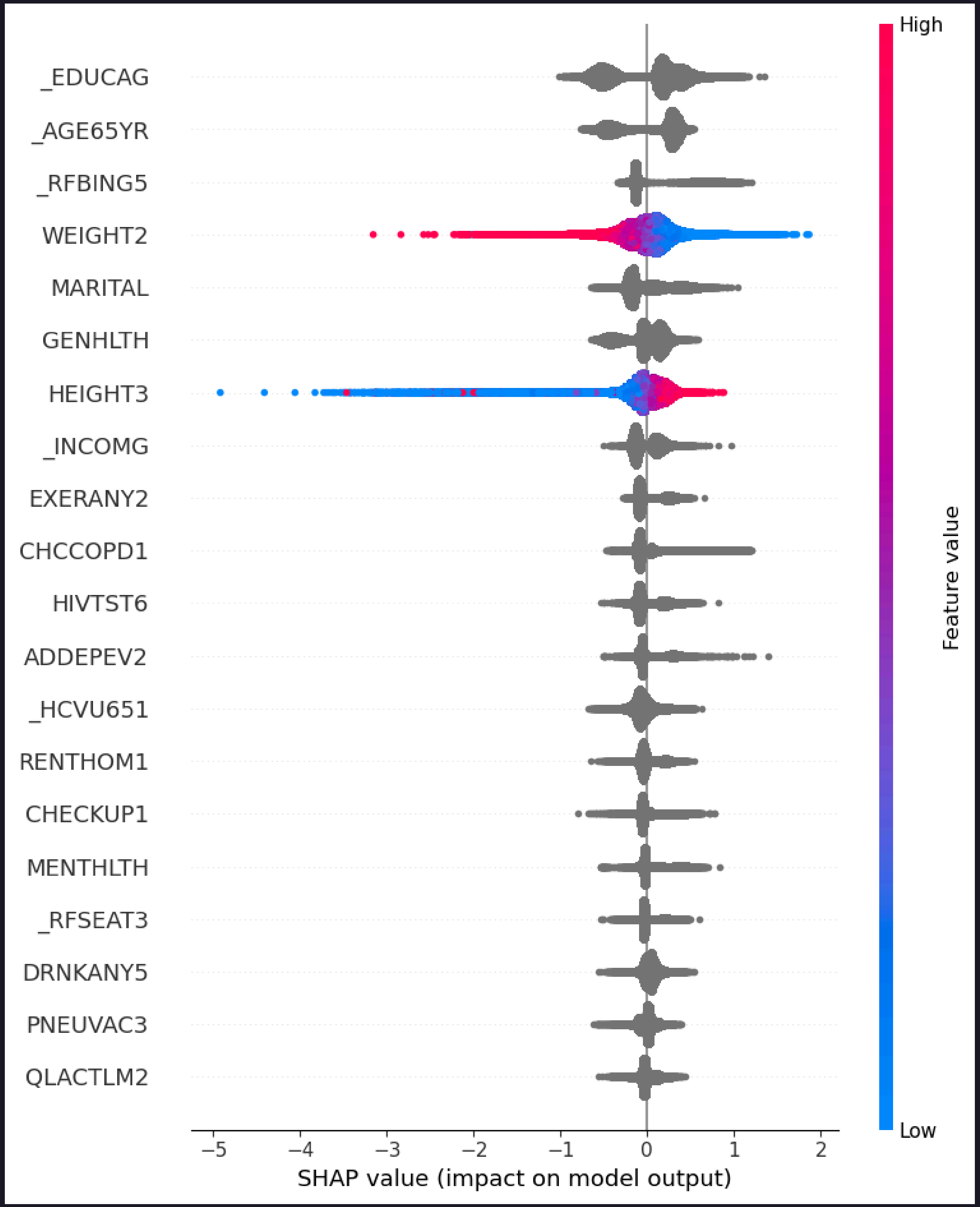
PART II -
Policy Recommendations

PRE-DATA WRANGLING

Pre-DataWrangling Steps to determine if we can use the current dataset to simulate various policy scenarios

1. Perform SHAP analysis on our prior model (CatBoost Model) to find out the most impactful levers guiding prediction of smoking
2. Perform SHAP analysis across age group



Outcomes

1. Our previous model has NaNs encoded either as 7 or 9, which needs to be handled to plot informative SHAP plots
2. Fix the order/encoding for some of the Ordinal variables to generate meaningful results on our SHAP plots
3. Label encode our variables to help in SHAP plots

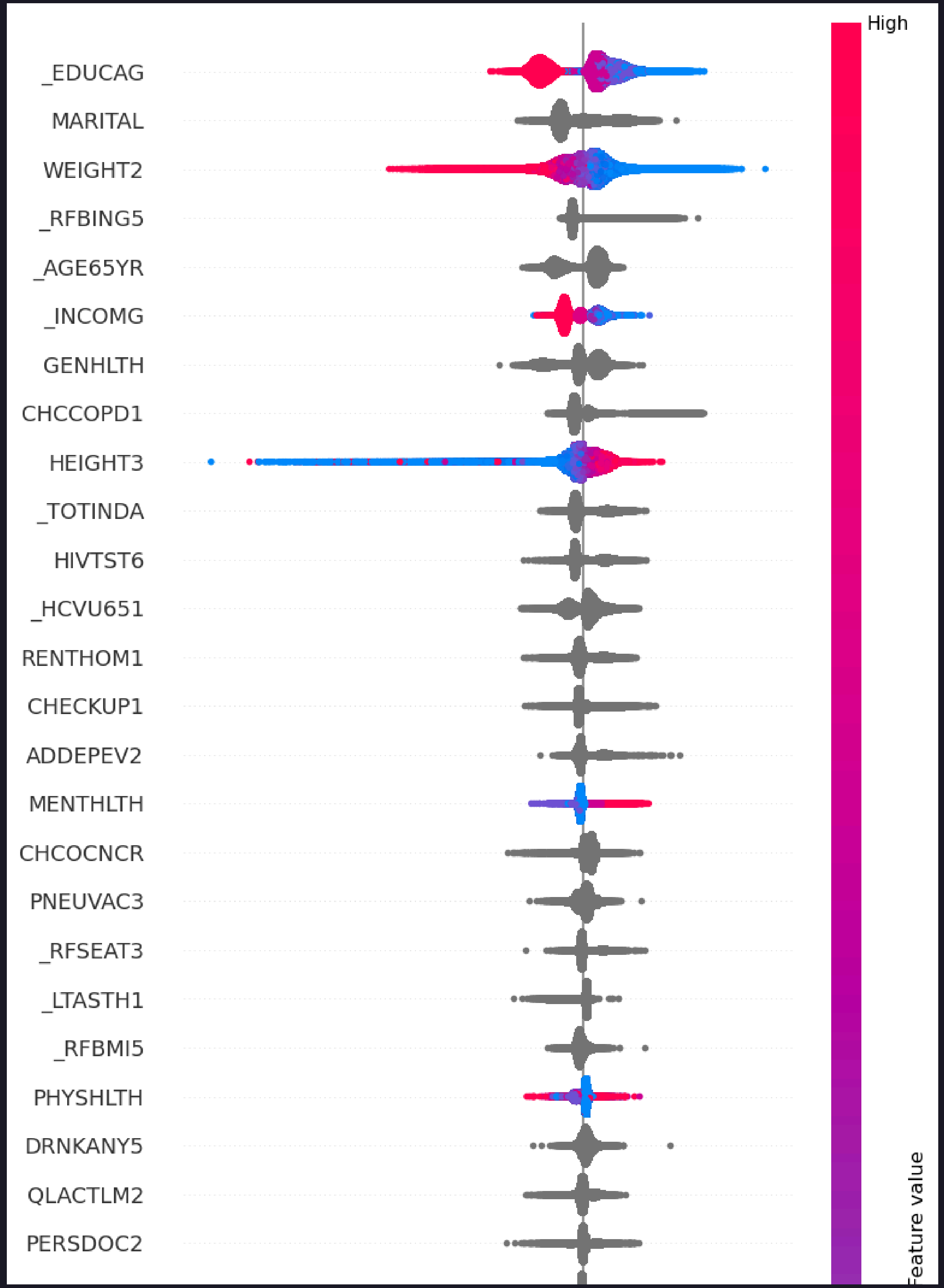
DATA WRANGLING

1. Large number of 7/9 values that don't given us any information.
2. Fix nominal ordering for categorical features like GENHLTH and _RFHLTH
3. Recode 7 to NaN, since Catboost Natively handles NaN values

EDA

Some tasks performed are -

1. Import new train/test data that has dealt with NaN
2. Import nominal/ordinal feature lists
3. Train new CatBoost Model for SHAP analysis on modified features
4. Train new CatBoost Model for SHAP analysis on modified features across age-groups



CONFOUNDER TEST

1. Check associations: 1.1. Each candidate variable ↔ _INCOMG.
   Each candidate variable ↔ _RFSMOK3.
2. Compare regression models: 2.1. Crude: smoking ~ income.
   Adjusted: smoking ~ income + candidate.

If the income coefficient changes ≥10%, candidate is a confounder.

Outcomes

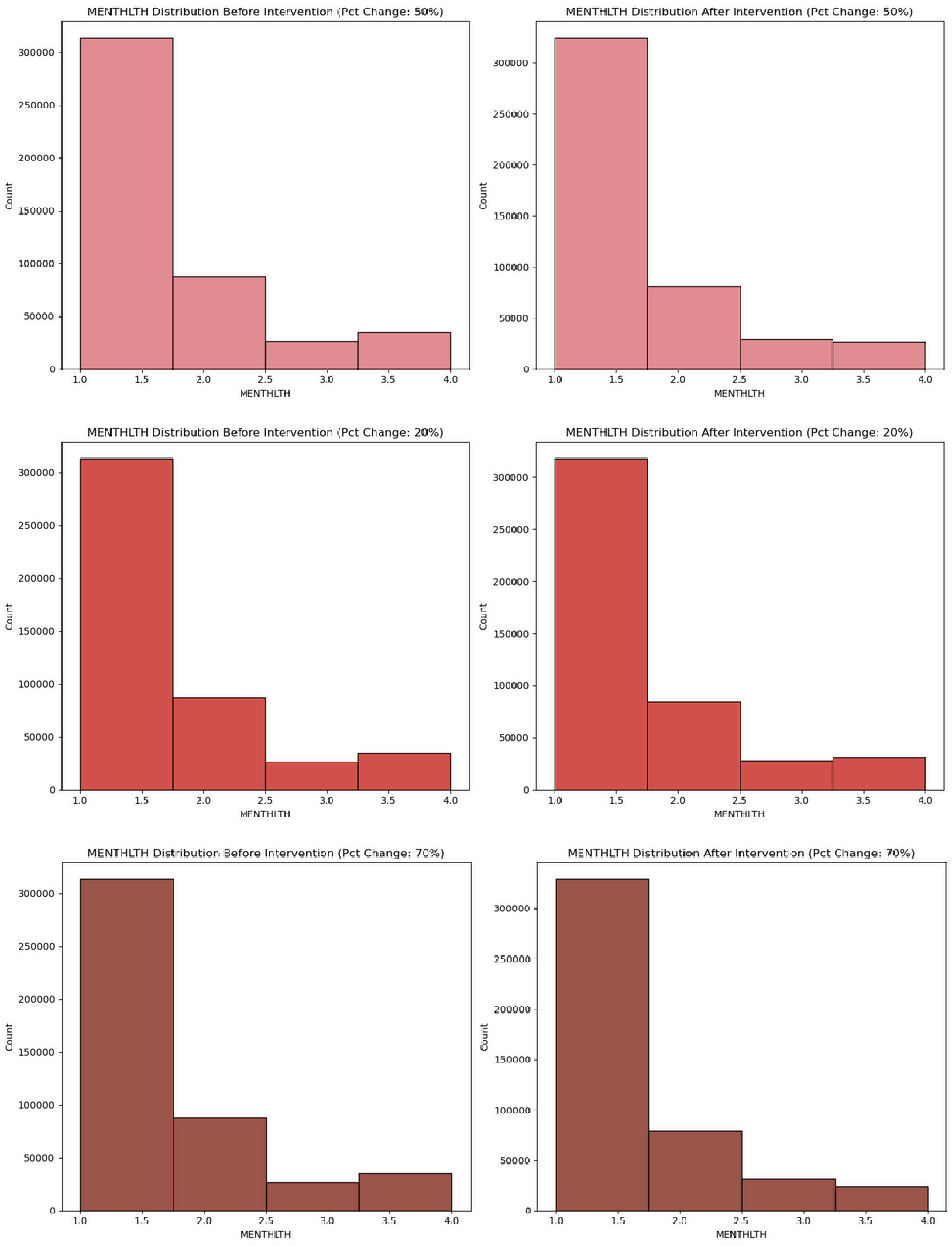
The money-related variables that represent crucial levers and barriers for smokers include:

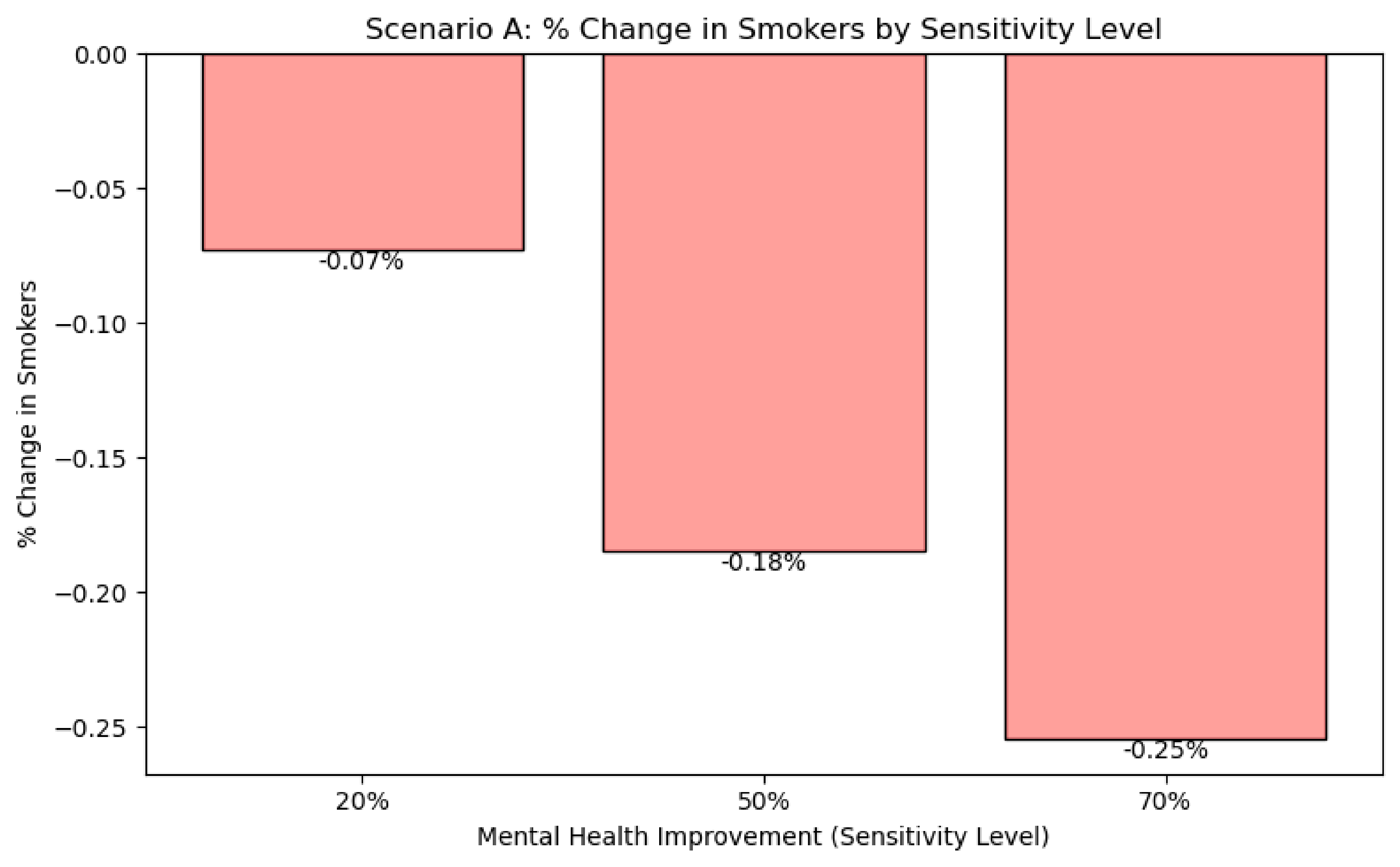
1. INCOME (_INCOMG) - Appears consistently across all age groups as a moderate to strong predictor, suggesting income level significantly impacts smoking behaviors/cessation success
2. Medical Costs (MEDCOST) - Shows up as a predictor across groups, indicating healthcare affordability affects smoking-related outcomes
3. Education (_EDUCAG) - While not directly money, education strongly correlates with earning potential and appears as a top predictor across all age groups
4. Healthcare Coverage (_HCVU651) - Insurance status appears as a factor, representing financial access to smoking cessation resources

Data Intervention Scenarios

Scenario A - Expand Mental Health Access

Goal: Reduce smoking by improving population mental health (MENTHLTH).
Rationale: MENTHLTH showed the largest confounding effect (up to ~28% change). Improving mental health should lower smoking propensity.
Intervention rule (synthetic):
For people in lowest two income groups (_INCOMG levels 1 & 2), reduce MENTHLTH measure by 20–30% (e.g. move categorical mental-health scores one level towards "better").
We shift 30% of respondents coded poor → fair, fair→good (stochastic selection within target group).
We also perform sensitivity analysis for % change in MENTHLTH by (20%, 50% and 70%) for lowest income groups 1 and 2
Metrics to report
expected_smokers_base and expected_smokers_sim (absolute counts)
Absolute and percent reduction
Stratified results for _INCOMG groups


MENTHLTH Distribution Before Intervention (Pct Change: 50%)
MENTHLTH Distribution After Intervention (Pct Change: 50%)
MENTHLTH Distribution Before Intervention (Pct Change: 20%)
MENTHLTH Distribution After Intervention (Pct Change: 20%)
MENTHLTH Distribution Before Intervention (Pct Change: 70%)
MENTHLTH Distribution After Intervention (Pct Change: 70%)
Count
MENTHLTH
0
50000
100000
150000
200000
250000
300000
1.0
1.5
2.0
2.5
3.0
3.5
4.0

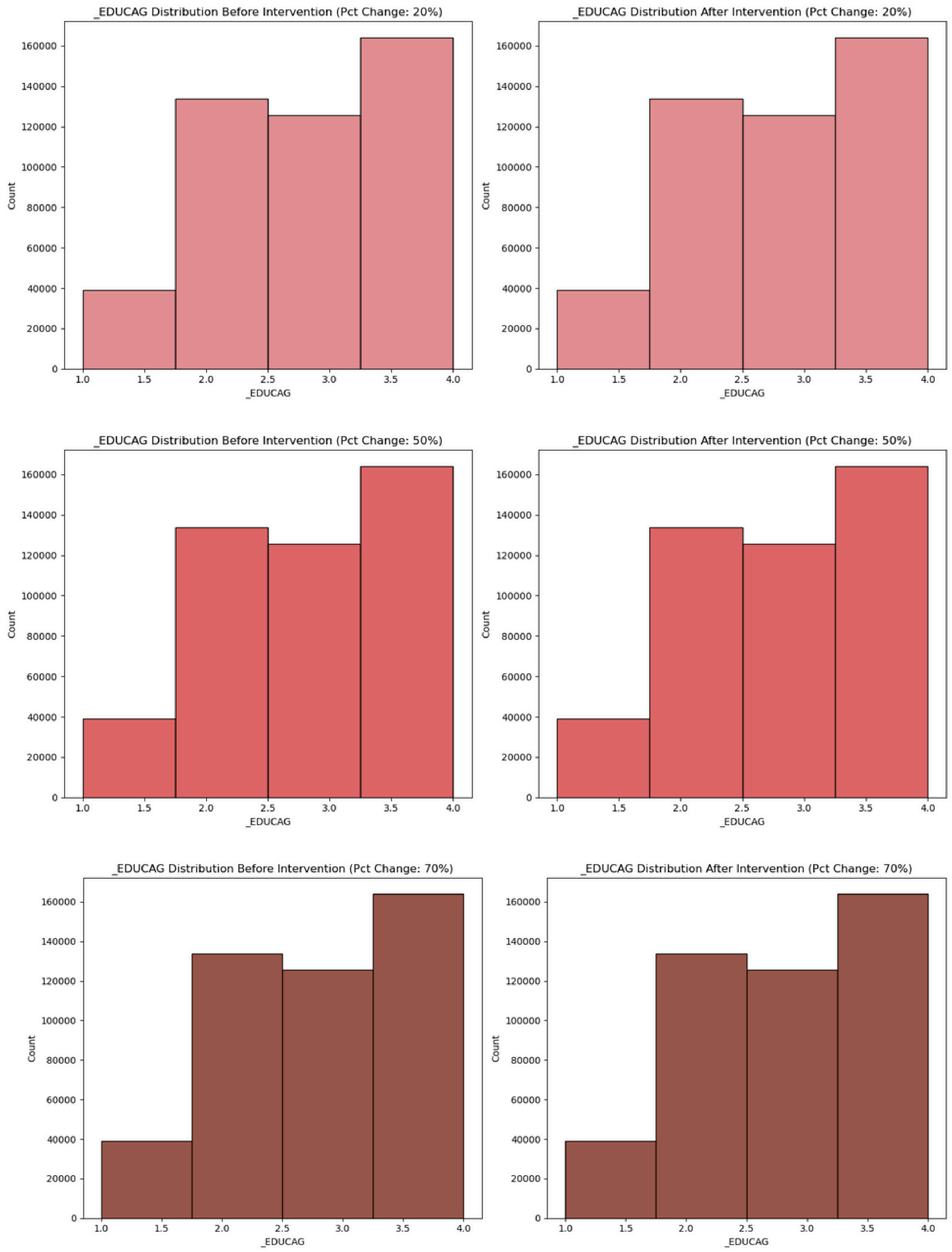


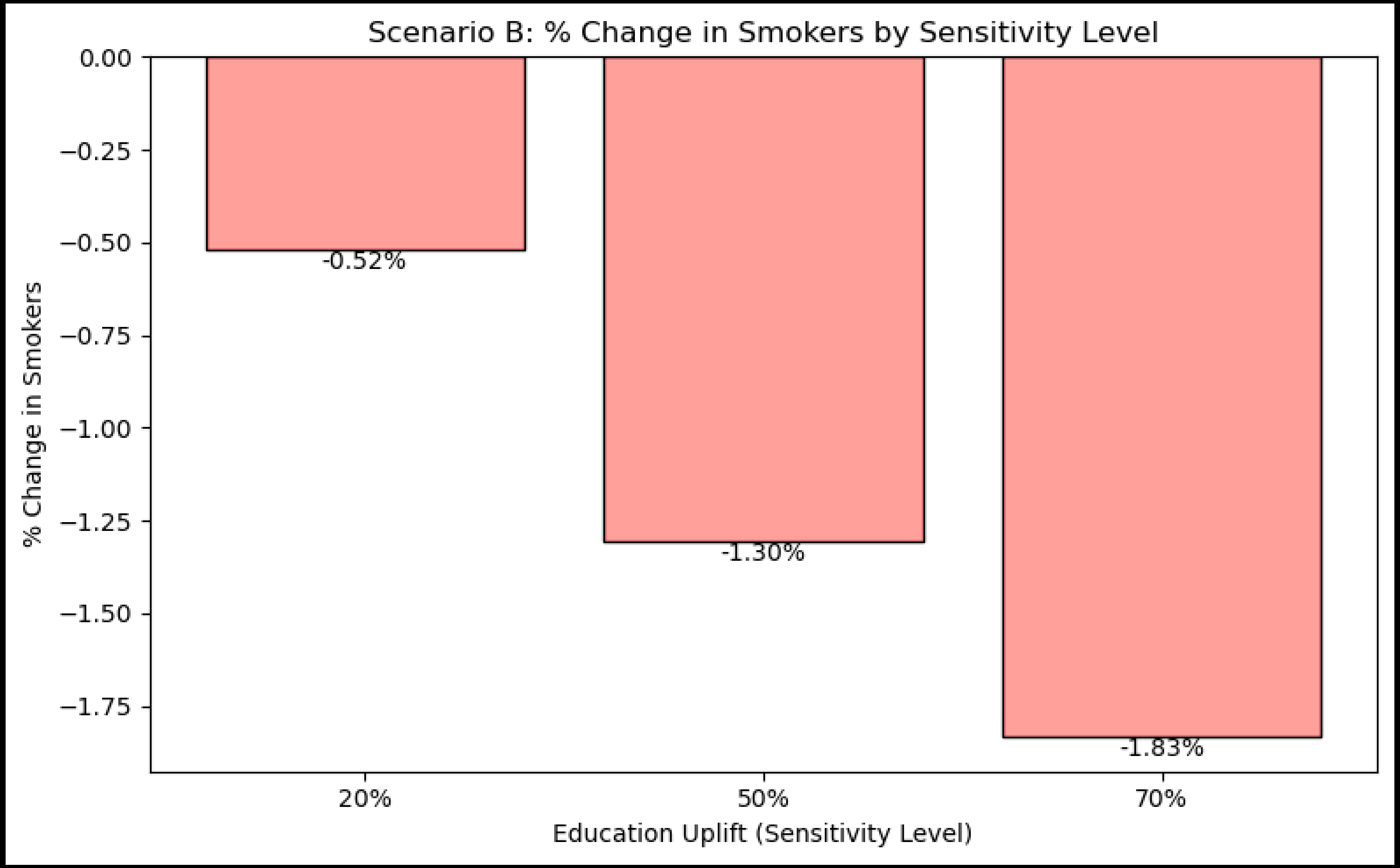
Interpretation

Mental health support may play a supportive role but is not sufficient by itself to drive large reductions in smoking.

Scenario B - Education Uplift Program

Goal: Simulate the impact of improving education (health literacy / adult education) in low-income groups.
Rationale: _ EDUCAG showed up to ~29% effect on the income coefficient — biggest lever found.
Intervention rule:
For _INCOMG = 1 & 2, upgrade one education level for 20% of respondents (e.g., Less than HS → High School or HS → Some College). This is a synthetic proxy for adult education / literacy programs.
Metrics & checks
Absolute and percent change in expected smokers overall and within _INCOMG groups.
Evaluate distributional effects: does benefit concentrate in certain ages?
Sensitivity: vary upgrade fraction (20%, 50%, 70%) and compare.


_EDUCAG Distribution Before Intervention (Pct Change: 20%)
_EDUCAG Distribution After Intervention (Pct Change: 20%)
_EDUCAG Distribution Before Intervention (Pct Change: 50%)
_EDUCAG Distribution After Intervention (Pct Change: 50%)
_EDUCAG Distribution Before Intervention (Pct Change: 70%)
_EDUCAG Distribution After Intervention (Pct Change: 70%)
Count
_EDUCAG
0
20000
40000
60000
80000
100000
120000
140000
160000
1.0
1.5
2.0
2.5
3.0
3.5
4.0



Interpretation

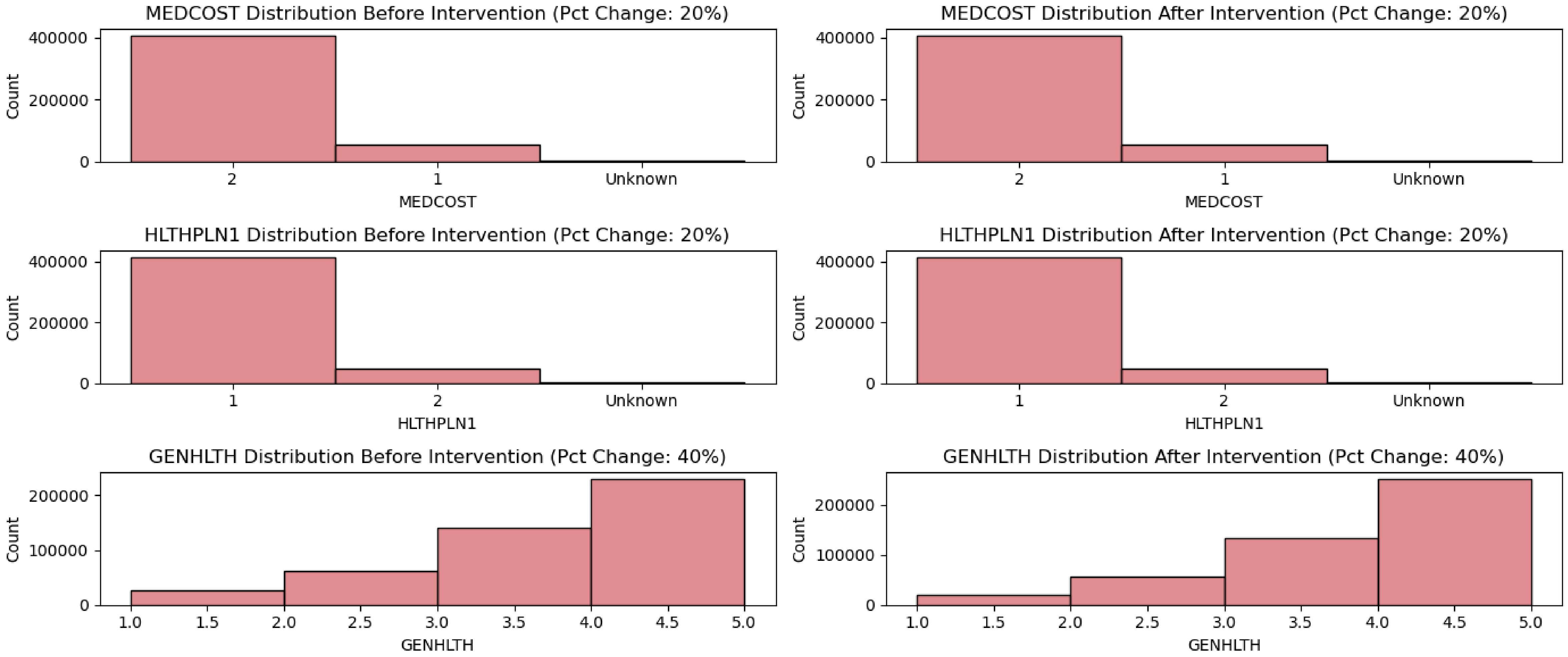
As individuals attain higher education, they are more aware of health risks and less likely to engage in smoking.

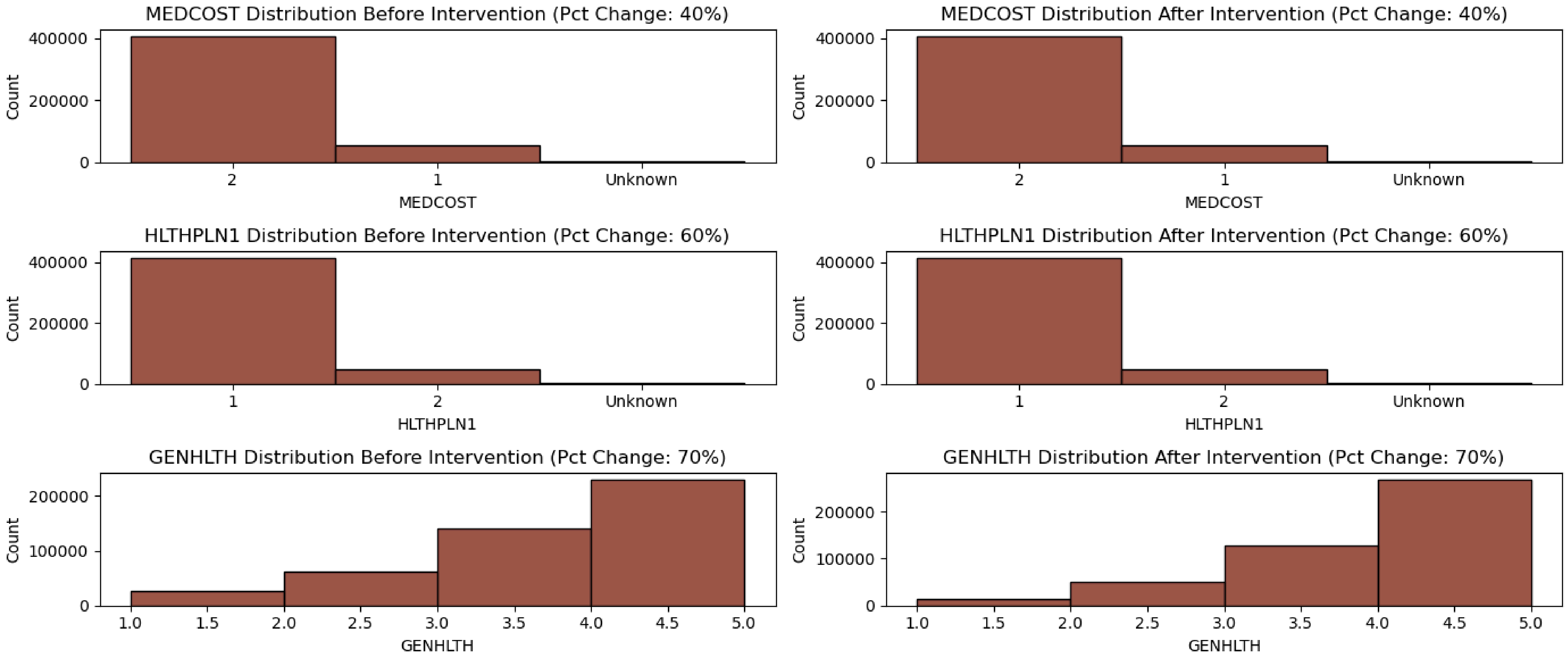
Scenario C - Remove Medical Cost Barriers + Insurance Outreach

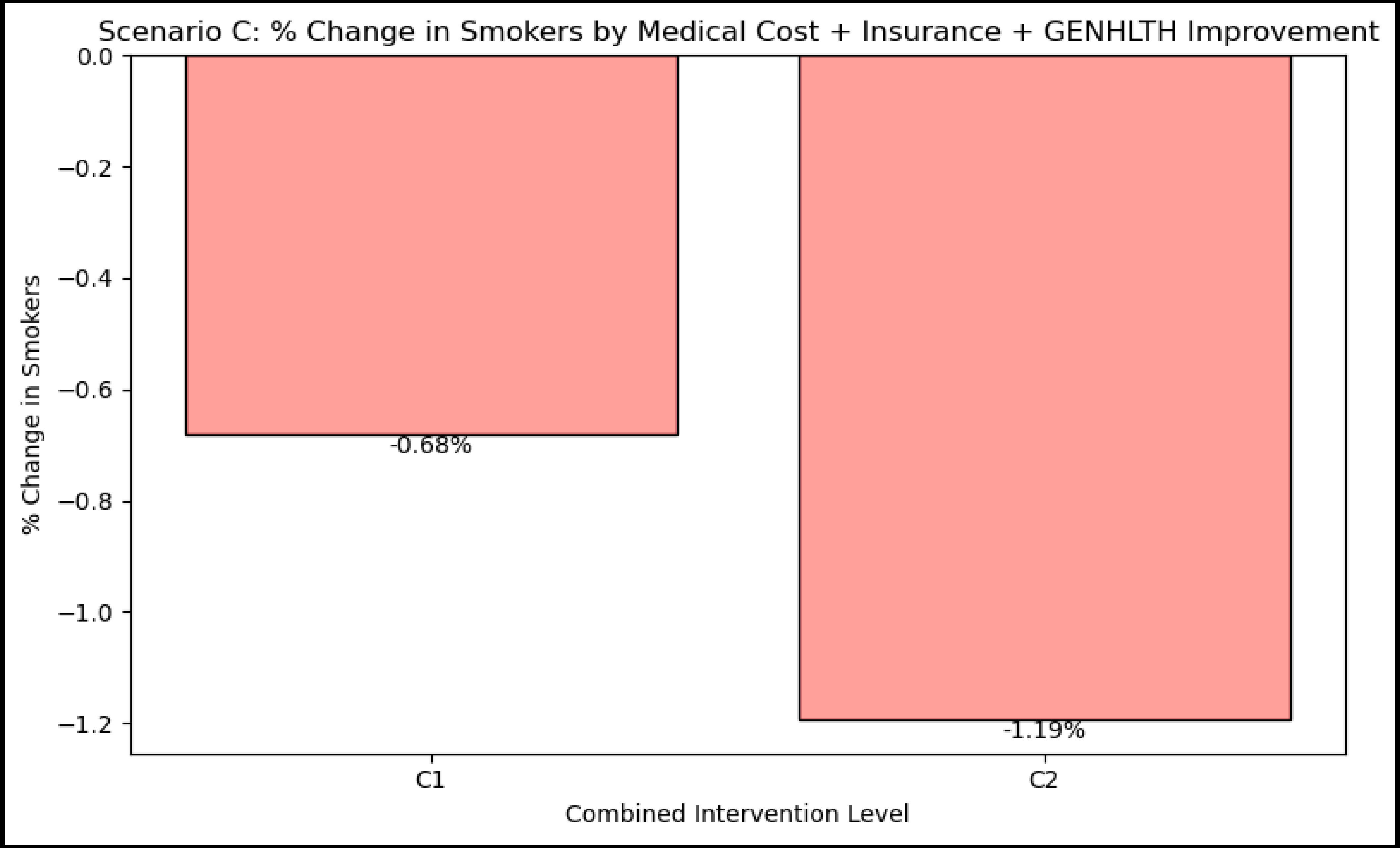
Goal: Reduce smoking by lowering reported medical cost barriers (MEDCOST) and increasing perceived insurance access (HLTHPLN1) in low/mid-income groups. We observe a improvement in GENHLTH which has also been modeled.
Rationale: MEDCOST , HLTHPLN1 and GENHLTH changed the income coefficient 11-21% — an actionable policy lever (subsidies, free cessation treatment).
Intervention rule:
For _INCOMG in [1,2,3],

1. Set MEDCOST -> No barrier for 20% of those who reported cost barrier. For same groups, flip HLTHPLN1 to insured for 20% (simulate outreach/coverage). GENHLTH as a result of improved MEDCOST and HLTHPLN1 by 40%. These ratios are picked up from the Confounder Analysis in EDA step.
2. Set MEDCOST -> No barrier for 20% of those who reported cost barrier. For same groups, flip HLTHPLN1 to insured for 30% (simulate outreach/coverage). GENHLTH as a result of improved MEDCOST and HLTHPLN1 by 50%. These ratios are picked up from the Confounder Analysis in EDA step.

Metrics
Expected smokers baseline vs simulated
Subgroup breakdown by income level and by whether they received benefit
Cost-effectiveness estimate (if you can attach program cost per person)


MEDCOST Distribution Before Intervention (Pct Change: 20%)
MEDCOST Distribution After Intervention (Pct Change: 20%)
Count
400000
200000
0
2
1
Unknown
MEDCOST
HLTHPLN1 Distribution Before Intervention (Pct Change: 20%)
HLTHPLN1 Distribution After Intervention (Pct Change: 20%)
HLTHPLN1
GENHLTH Distribution Before Intervention (Pct Change: 40%)
GENHLTH Distribution After Intervention (Pct Change: 40%)
100000
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
GENHLTH


MEDCOST Distribution Before Intervention (Pct Change: 40%)
MEDCOST Distribution After Intervention (Pct Change: 40%)
Count
400000
200000
0
2
1
Unknown
MEDCOST
HLTHPLN1 Distribution Before Intervention (Pct Change: 60%)
HLTHPLN1 Distribution After Intervention (Pct Change: 60%)
HLTHPLN1
GENHLTH Distribution Before Intervention (Pct Change: 70%)
GENHLTH Distribution After Intervention (Pct Change: 70%)
100000
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
GENHLTH



Interpretation

Reduced medical costs ease financial barriers to care, expanded insurance improves access to cessation programs, and better general health reflects increased well-being and readiness to quit.